

1> Why is Data Communication Important?..

→ Data Communication is the electronic transfer of information (Such as text, audio, or video) between a sender and receiver via a transmission medium.

→ Importance of Data Communication:-

1> Fast Communication.

- Data communication allows information to be transmitted quickly over short and long distances.
- Messages, files, audio & video can be shared instantly.

2> Resource Sharing:-

- User can share hardware, software and data resources such as printers, storage devices, and internet connection through networks.

3> Accuracy and Reliability.

- Digital communication costs by enabling electronic transfer of information instead of physical transportation of documents.

4> Cost efficiency:-

- It decreased communication costs by enabling electronic transfer of information instead of physical transportation of documents.

5> Remote Access.

- People can access systems, databases, & services from remote locations using communication networks.

6) Supports Business operations:-

- Banks, schools, hospitals, industries, and organizations depend on data communication for daily operations, online transaction and management systems.

7) Global connectivity

- Data communication connects people worldwide through the internet, making global communication easy and effective.

2. Draw and Explain a Generic Data Communication Model. Explain in brief the basic signal processing operations at each block

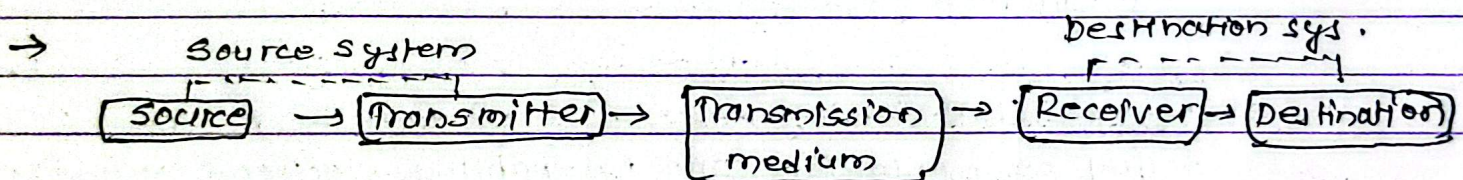


Fig:- Block diagram of data communication system.

a) Source:-

- The source generates the information or data that will be transmitted to the destination.
- popular form of information include text, number, picture, audio, video or combination of any of these.

b) Transmitter

- The transmitter is a device which is used to convert the data as per the destination requirement.
- e.g:- A modem (Modulator & Demodulator) convert the analog (Telephone) signal to digital computer signal and alternatively digital to analog.

c) Transmission Medium:-

- The transmission medium is the physical path by which data travels from transmitter to receiver.

d) Receiver:-

- The receiver receives the signal from the transmission medium and convert it into a form that is suitable to the destination device.

⇒ For e.g:- a modem accepts analog signal from a transmission channel and transforms it into digital signal which is acceptable by computer system.

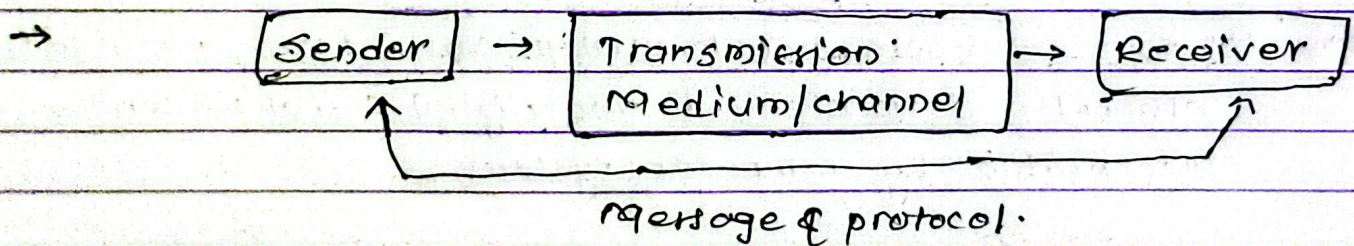
e) Destination:-

- The destination takes the incoming data from the receiver.

⑧ → Working process of Generic Data Communication Model.

- 1) The sender generates information.
- 2) The transmitter processes and converts the data into signals.
- 3) The communication channel carries the signals.
- 4) The receiver reconstructs the original data from received signals.
- 5) The destination receives and uses the information.

⑨ Explain the five Essential components of a Data Communication system. How does each components contribute to successful communication? Explain with suitable diagram.



→ A data communication system requires five basic components for successful transmission of data from one device to another.

1) Message:-

- The message is the actual information or data to be communicated from sender to receiver.

eg:- Text, Audio, video, Images, Numerical data.

- Contribution to communication:-

- It is the main content being transmitted.
- Without a message communication cannot occur.

2> Sender .

- The sender is the device or the person that sends the message.
- eg:- computer, mobile phone, sensor, TV station.

- Contribution to communication:-

- Generates & transmits the data.
- Converts information into ~~strong~~ signals suitable for transmission.

3> Receiver

- The Receiver is the device or person that receives the message.
- eg:- Computer, Smartphone, printer, TV.

- Contribution to Communication:-

- Accepts and interprets the received signals.
- converts signal back into understandable information.

4> Transmission Medium .

- Transmission medium is the physical or wireless path through which data travels from sender to receiver.

eg:- Twisted pair cable, coaxial cable, optical fiber,
Radio waves, satellite communication.

- contribution to communication.

- provides a path for signal transmission.
- Determines speed, reliability, & quality of communication.

5> protocol.

- A protocol is a set of rules that governs data communication.

eg:-

TCP/IP, HTTP, FTP, SMTP.

- contribution to communication:-

- Ensures proper formatting and synchronization of data.
- Controls error handling, flow control, & transmission procedure.
- Enables devices from different manufacturers to communicate successfully.

Q. How is Digital Transmission Superior to Analog Transmission System?

→ Digital Transmission is considered superior to analog transmission because it provides better, quality, higher reliability, improved security and efficient data handling. In modern communication systems, digital transmission is widely used in computers, mobile networks, and the Internet.

⇒ Difference between Digital And Analog Transmission.

Basis	Digital Transmission	Analog Transmission.
1) Signal type	- use discrete binary signals (0 & 1)	- uses continuous signals
2) Noise effect	- less affected by noise	- highly affected by noise.
3) Signal quality	- maintains quality over long distance	- Quality decrease with distance.
4) Error Detection	- Easy error detection and correction	- Difficult error detection.
5) Data Security	- More secure using encryption	- less secure.
6) Storage & processing	- Easy to store and process	- Difficult to process.
7) Bandwidth utilization	- Efficient bandwidth usage	- less efficient.
8) Reliability	- Highly reliable	- comparatively less reliable.
9) Multiplexing	- Easy implementation	- more complex.
10) eg:-	- computer network, digital TV	- AM/FM radio, etc

⇒ Advantage of digital Transmission Over Analog Transmission.

1> Better noise Immunity

- Digital signals are less affected by external noise and interference. Even if noise is added, the original data can be recovered accurately.

2> Error Detection and correction.

- Digital System uses error control techniques to detect and correct transmission errors.
- eg:- parity checking, CRC, hamming code

3> Higher Data Security

- Digital data can be encrypted easily, making communication more secure.

4> Easy storage and processing:-

- Digital data can be stored, compressed, edited, and processed easily using computers and digital devices.

5> Better Signal Quality Over long distance.

- In digital communication, repeaters regenerate the original signal, preventing quality loss over long distances.

- In analog systems, amplifiers also amplify noise.

6> Supports Modern Technologies.

7> Efficient Multiplexing:-

- Multiple digital signal can be transmitted together efficiently using technique like:- Time division multiplexing.

⑤ Explain Digital Communication System with its Block diagram.

→ A digital communication system is a system that transfers information from one place to another in the form of digital signals. It ensures reliable, fast, and efficient communication.

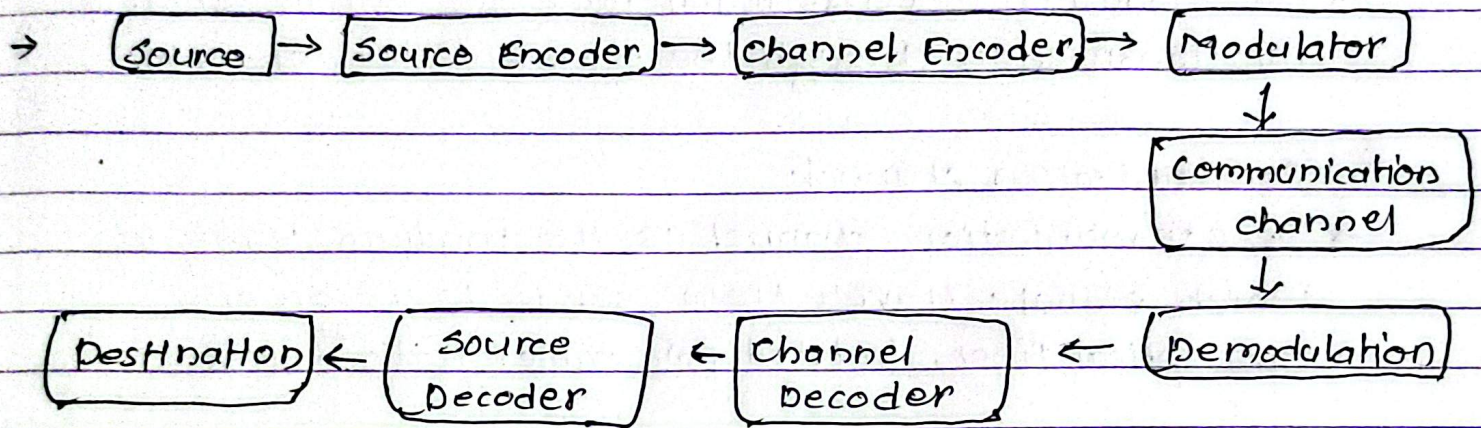


Fig:- Block Diagram of Digital Communication System.

→ Explanation of Each Block.

① Information Source.

- The information source generates the original message or data.
- eg:- text, audio, video, image.

② Source Encoder

- The source encoder converts the original information into digital binary form and removes unnecessary redundancy.

③ Channel Encoder.

- The channel encoder adds extra bits to the data for error detection and correction.
- eg:- Hamming codes, parity bits.

④ Modulator

- The modulator converts digital data into signals suitable for transmission through the communication channel.

⑤ Communication channel.

- The communication channel is the medium through which signals travel from sender to receiver.
- eg:- optical fiber, Twisted pair cable, Radio waves,

⑥ Demodulator.

- The demodulator extracts the original digital data from the received signal.

⑦ Channel Decoder

- The channel decoder detects and corrects errors introduced during transmission.

⑧ Source Decoder

- The source decoder reconstructs the original message from digital data.

⑨ Destination:-

- The destination is the final user or device that receives the information.

⑥ Explain the Applications of Data Communication.

→ Data Communication is used in many fields to exchange information between devices and user efficiently. It plays an important role in modern society by enabling fast, reliable and secure communication.

→ Application of Data Communication.

1. > Business Communication.

- Organizations use data communication for sharing information online meetings, file transfer & business transactions.

2. > Education.

- Educational institutions use data communication for online learning and sharing educational resources.

3. > Banking and Finance.

- Banks use data communication for secure financial transactions and customer services.

4. > Healthcare

- Hospitals and healthcare systems use communication n/w's to manage patients information & provide remote medical services.

5. > Internet Services.

- The internet is one of the largest applications of data communication.

6> Industrial Automation

- Industries use data communication for monitoring & controlling machines and production system.

7> Entertainment.

- Data Communication supports digital entertainment services like online gaming, video streaming etc.

Chapter:- 2.

① Define Baud Rate. Explain RS-232C standards.

⇒ Baud Rate is the number of signal changes or symbols transmitted per second in a communication channel.

→ It indicates how many times the signal changes its state in one second.

→ Formula:-
$$\text{Baud Rate} = \frac{\text{Bit Rate}}{\text{No of bit per signal element}}$$

→ When one signal element carries one bit

$$\text{Baud Rate} = \text{Bit Rate}$$

⇒ RS-232C Standard.

→ Recommended Standard - 232C is a standard used for serial communication between computer and peripheral devices.

→ It was developed by the electronic Industries Association.

→ It defines:-

- Electrical signal characteristics.
- voltage levels.
- Timing of signals.
- Connector types.
- Communication protocol for serial data transfer.

→ features of RS-232C.

1> Serial Communication standard:-

- Data is transmitted one bit at a time through a single communication line.

2> Used for short distance communication.

- Typically used for communication within short distance (15 meter) (50 feet)
- eg:- Computer to modem.
Computer to printer
Computer to mouse.

3> Voltage levels.

Logic	Voltage Range
Binary 1 (mark)	-8V to -25V
Binary 0 (space)	+8V to +25V

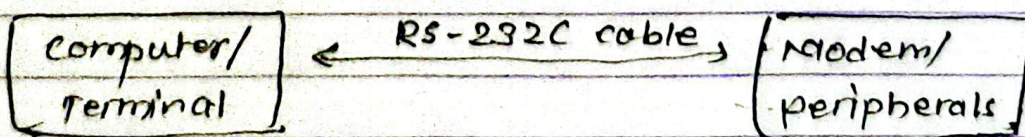
- Voltage between -8V & +8V are considered undefined.

4> Connector type

RS-232 commonly uses:-

- DB-25 connector
- DB-9 connector.

5> It supports simultaneous two way communication.



6. → Important signals in RS-232C.

Signal	Full form	Function
TXD	Transmit Data	Send Data
RXD	Receive Data	Receives Data
RTS	Request to send	Request for transmission.
CTS	clear to send	permission to send.
DTR	Data terminal Ready	Terminal ready signal
DSR	Data set Ready	Device ready signal
GND	Ground	Reference voltage.

② Explain data frame format of Synchronous and Asynchronous communication and Different Them.

→ Data transmission in communication system can be preferred performed using two methods:-

1. Asynchronous Communication.

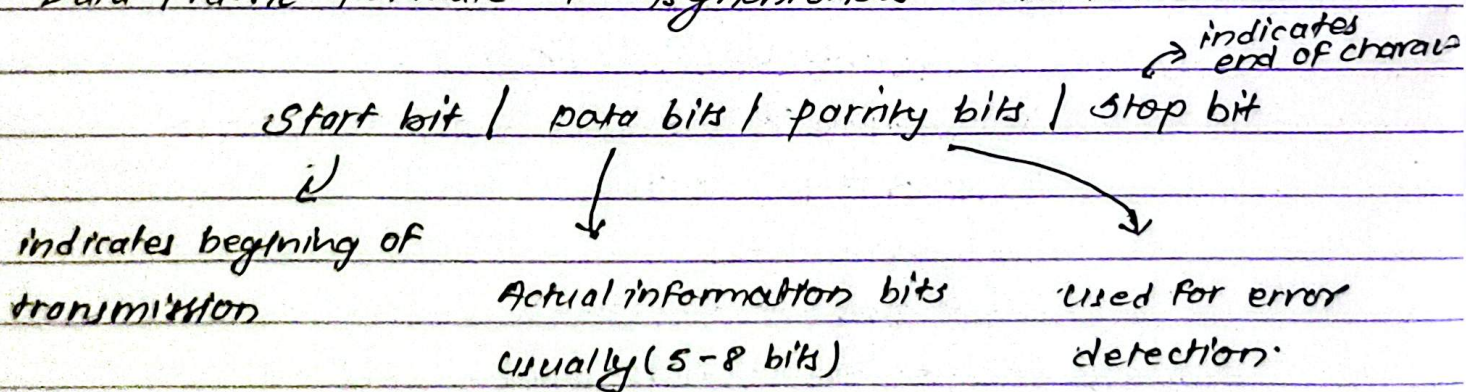
2. Synchronous Communication.

→ Both methods use different frame format for transmitting data.

Q. Asynchronous Communication:-

→ In asynchronous communication, data is transferred character by character independently. Each character contains additional bit for synchronization.

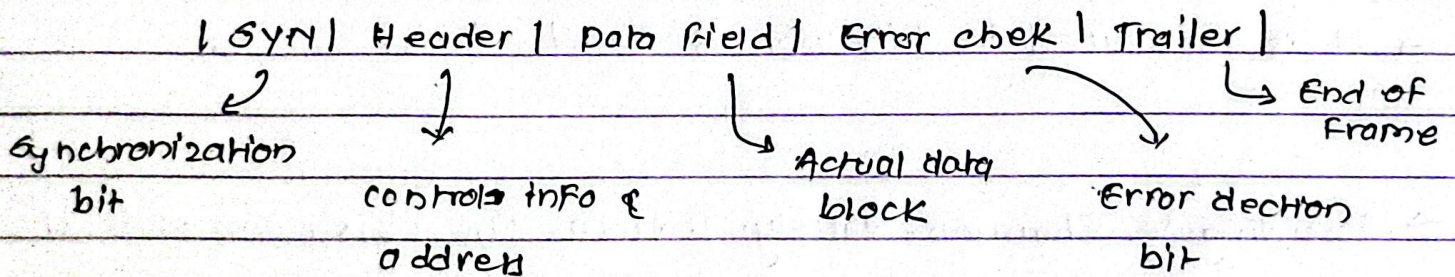
→ Data frame format of Asynchronous Communication.



b) Synchronous Communication:-

- In synchronous communication, data is transmitted in block or frames continuously without start & stop bits for each character.
- Both sender and receiver use synchronized clocks.

→ Data frame Format of Synchronous communication:-



③ Define channel capacity with necessary Theorem. for a RF signal with Bandwidth 6 MHz & it is transmitted with Average power 2 W where 1.2 mW Noise signal is present on it. If the signal is digitized & sent, find the Maximum Data Rate of the channel.

⇒ Channel capacity is the maximum rate at which data can be transmitted through a communication channel reliably without error.

→ It is measured in bits per second (bps)

→ According to Shannon-Hartley Theorem, the maximum channel capacity is

$$C = B \log_2(1 + \text{SNR})$$

where,

C = channel capacity (bps)

B = Bandwidth of channel (Hz)

SNR = Signal to noise - Ratio.

⇒ Given data,

$$B = 6 \text{ MHz} = 6 \times 10^6 \text{ Hz}$$

$$S = 2 \text{ W}$$

$$N = 1.2 \text{ mW} = 1.2 \times 10^{-3} \text{ W}$$

now,

$$\text{SNR} = \frac{S}{N} = \frac{2}{1.2 \times 10^{-3}} = 1666.67$$

Applying Shannon Capacity Formula.

$$C = 6 \times 10^6 \cdot \log_2(1 + 1666.67)$$

$$= 6 \times 10^6 \times 10.7$$

$$= 64.2 \times 10^6 \text{ bps.}$$

$$\therefore C = 64.2 \text{ Mbps.}$$

Q. Explain different modes of serial transmission:-

- Serial transmission is a method in which data bits are transmitted one bit at a time through a ~~single~~ single communication channel.
- The different modes of serial transmission are classified based on synchronization between sender and receiver.
- Modes of serial transmission:
 - 1. → Asynchronous transmission.
 - 2. → Synchronous .11
 - 3. → Isochronous .11.

Q. Asynchronous Transmission:-

- Data is transmitted character by character independently.
- Each character contains start and stop bits for synchronization.

| Start | Data bits | Stop |

- Features:-

- No common clock required.
- Simple & Inexpensive
- Data transmitted at irregular intervals.
- More overhead due to start & stop bits.

DisAdvantages:-

- Slower transmission
- less efficient

~~Dis~~Advantage

- ~~Slower transmission~~
- ~~less e~~

Advantage:-

- Easy Implementation.
- low cost
- Suitable for low speed communication.

Application:-

- Keyboard communication
- Serial port.
- RS-232 communication.

(b) Synchronous transmission.

- In synchronous transmission, data is transmitted continuously in blocks or frames.
- Sender and Receiver share synchronized clocks.

[SYN | Header | Data block | CRC]

• Features:-

- Continuous data transmission
- No start & stop bits for each character.
- High transmission speed.
- Efficient for large data transfer.

• Advantage

- Faster communication
- Better efficiency
- Suitable for high speed n/w

• Disadvantage.

- More complex
- Requires synchronization mechanism.

• Application:-

- Computer networks.
- Digital communication system.
- Data center.

© Isochronous Transmission.

- In Isochronous transmission, data is transmitted at fixed time intervals.
- It ensures constant delivery time for real time applications.

• Features:-

- Fixed timing for data delivery
- Minimal delay variation.
- Suitable for real time communication.

• Advantages

- Smooth real time transmission.
- constant data rate
- Reduced delay.

• Disadvantage

- complex timing control.
- Expensive Implementation.

• Applications.

- Audio streaming
- video conferencing
- multimedia system.

⇒ Difference b/w:-

Bits	Asynchronous	Synchronous	Isochronous.
Synchronization	Start/stop bit	common clock	Fixed Timing
Data transfer	character-wise	Block-wise	Real time contin.
Speed	slow	fast	constant
Efficiency	low	high	high.
complexity	simple	moderate	complex
Application	keyboard RS-232	Networks	Audio/Video streaming.

⑤ Explain the concept of bit Rate and Baud Rate. Show their Relationship.

- In DC, bit rate and baud rate are important terms used to measure transmission speed.
- Although they are related, they are not always the same.

1. Bit Rate

- Bit Rate is the number of bits transmitted per second through a communication channel.
- It measures the actual amount of data transferred.

- formula:-

$$\text{Bit Rate} = \frac{\text{No of bits transmitted}}{\text{Time}}$$

unit = bits per second (bps)

2. Baud Rate

- Baud Rate is the number of signal changes or symbols transmitted per second.
- It represents how many times the signal changes state in one second.

- formula:-

$$\text{Baud Rate} = \frac{\text{Number of signal elements}}{\text{Second}}$$

unit:- Baud.

⇒ Relationship between Bit Rate and Baud Rate.

- The Relationship depends on the number of bits represented by each signal element.

~ formula:-

$$\text{Bit Rate} = \text{Baud Rate} \times \text{Bits per signal element.}$$

or,

$$\text{Baud Rate} = \frac{\text{Bit Rate}}{\text{Bits per signal element.}}$$



If one signal element carries one bit.

$$\text{Bit rate} = \text{Baud Rate.}$$

If one signal element carries multiple bits.

$$\text{Bit Rate} > \text{Baud Rate.}$$

⑥ If a communication channel has a Bandwidth of 8 kHz and SNR of 81 dB, what would be the channel capacity using Shannon capacity theorem?

→ According to Shannon capacity theorem,

$$C = B \cdot \log_2 (1 + \text{SNR})$$

Given data,

$$B = 8 \text{ kHz} = 8000 \text{ Hz}$$

$$\text{SNR} = 81 \text{ dB}$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} (\text{SNR})$$

$$\frac{81}{10} = \log_{10} (\text{SNR})$$

$$8.1 = \log_{10} (\text{SNR})$$

$$\therefore \text{SNR} = 1258.92$$

Apply Shannon capacity Formula,

$$C = 8000 \log_2 (1 + 1258.92)$$

$$C = 80900 \text{ bps}$$

$$\therefore C = 80.9 \text{ kbps}$$

⑦ Explain Transmission Modes in Data Communication:

→ Transmission Modes defines the method and direction in which data is transferred between two devices in a communication system.

⇒ Transmission modes in data communication can be classified based on:-

1. Direction of data flow

2. Number of bits transmitted at a time.

3. Synchronization between devices.

① Transmission Modes Based on Direction of Flow:-

① Simplex mode

- Communication occurs in only one direction.

- One device acts only as sender & other only as receiver.

Sender → receiver.

- Features:-

- One way communication

- No feedback possible.

- Simple communication system.

- eg:- Radio broadcasting,

Television II

Keyboard to computer.

⑥ Half Duplex Mode

- Communication occurs in both directions but not simultaneously.
- Both devices can send and receive data one at a time.

Device A $\longleftrightarrow$ Device B
One direction at a time.

- Features:-

- Two way communication.
- Only one device transmits at a time.
- Requires coordination.

- eg:- walkie-talkie, CB radio.

⑦ Full-Duplex Mode

- Communication occurs simultaneously in both directions.
- Both devices can send and receive data at the same time.

- Device A $\longleftrightarrow$ Device B
(Simultaneous communication).

- Features:-

- Simultaneous two way communication.
- Faster & more efficient.
- Better channel utilization.

- eg:- telephone, mobile communication, video conferencing.

② Transmission Modes Based on Number of bits:

① Serial Transmission:-

- Data bits are sent one bit at a time through a single channel.

- Features:-

- low cost
- Reliable for long distance.
- less interference.
- eg:- USB, ethernet

② Parallel Transmission:-

- Multiple bits are transmitted simultaneously using multiple channels.

- Features:-

- faster for short distance.
- More expensive
- Higher interference.
- eg:- printer ports, Internal buses.

⑧ Transmission Modes Based on Synchronization.

① Asynchronous Transmission.

- data is transmitted character by character using start & stop bits.

- Features:-

- No common clock required.
- Simple implementation.

- eg:- keyboard communication
RS-232

② Synchronous Transmission

- data is transmitted continuously in blocks using synchronized clocks.

- Features:-

- High speed
- Efficient transmission.

- eg:- Computer N/w, data communication system.

③ Isochronous Transmission.

- data is transmitted at fixed time interval for real time communication.

- Features:-

- constant data delivery rate
- minimal delay

- eg:- Audio/video streaming, Multimedia.